APPENDIX
I. TEST CASES AND SIMULATION PARAMETERS

A. Test Cases

The large-scale dispatch case uses a balanced AC 69-bus radial feeder with constant-power loads. Line impedances, base loads, and branch current ratings follow the public dataset in [1]. Fig. A1 shows the feeder and microgrid connections: each of nine installation buses hosts 100 PCC–microgrid pairs, yielding 900 PCCs and 2,700 resource units. The model contains 10,800 continuous controls and 900 binary reactive-support modes. PCC exchanges are coupled through feeder voltages, branch currents, and losses.

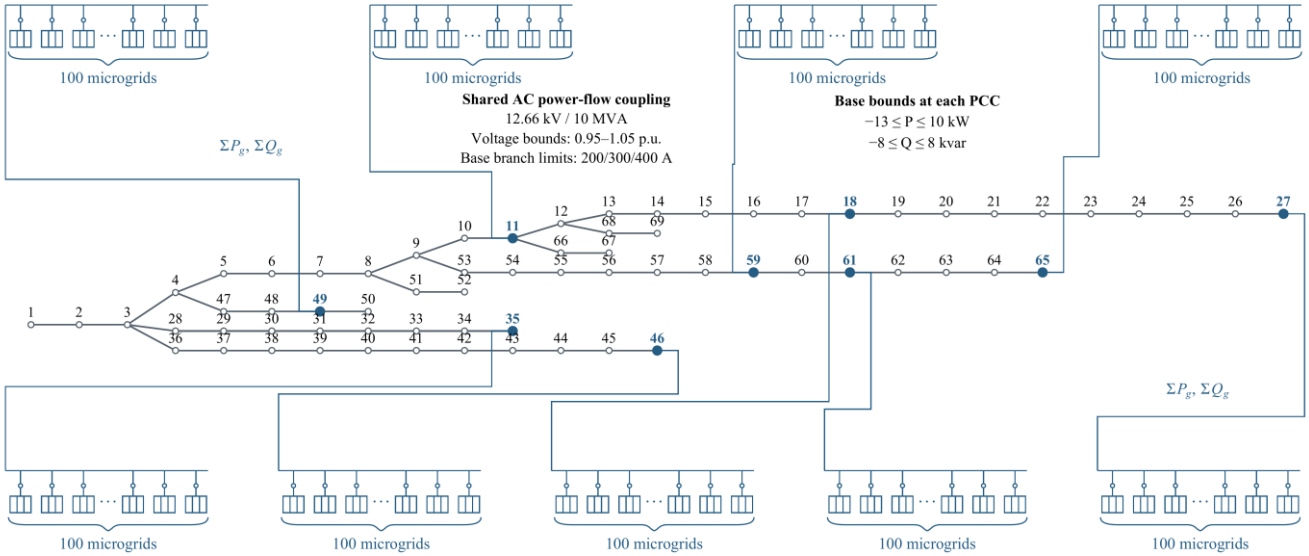


Fig. A1. Integrated 900-PCC test system. Aggregate PCC exchanges couple the microgrids through shared AC power flow, voltage bounds, and branch current limits.

B. Simulation Parameters

The network uses 12.66 kV and 10 MVA bases, with bus 1 fixed at 1.00∠0° p.u. Aggregate resource ratings are allocated among the microgrids at each installation bus. Table A1 lists the feeder and resource settings. Operating profiles use DE_KN_residential3 in the OPSD Household Data package [2].

TABLE A1 FEEDER AND RESOURCE PARAMETERS

Parameter	69-bus, 900-PCC configuration
Radial branches / installation buses	68 / 9
Base active / reactive feeder load	3.80189 MW / 2.69410 Mvar
Base branch current limits	200 / 300 / 400 A, as specified by branch [1]
Aggregate PV active power rating	8.0 MW
Aggregate storage power / energy rating	2.8 MW / 7.0 MWh
Aggregate reactive support / load-shedding rating	3.4 Mvar / 0.8 MW
Additional aggregate microgrid load base	1.2 MW
Voltage limits / dispatch interval	0.95–1.05 p.u. / 0.25 h
Storage SOC limits; charge / discharge efficiency	0.20–0.90; 0.95 / 0.95
PV inverter apparent power rating	1.10 × rated PV active power
Base exchange bounds per PCC	[−13, 10] kW active; [−8, 8] kvar reactive

The LLM compiles operating requirements into typed Rules and coordinates Skills through three parallel strategy subagents. The service is deepseek-flash (temperature 0.2, thinking disabled); rule and strategy output limits are 1200 and 600 tokens. Numerical execution uses one thread and a 90 s budget. Sparse elastic SQP allows 50 iterations; Clarabel allows 150 iterations with tolerance 10^{-9}. Independent AC power-balance verification uses a 10^{-8} p.u. tolerance. Device, SOC, and independently checked task constraints use 10^{-8} in their respective scales; normalized network inequalities use 10^{-7}.

Initial SOC is sampled in 0.42–0.75 (0.205 for restricted units). Base cost weights are 0.8–1.2 for storage, 0.04 for reactive power, 5–7 for load shedding, and 4 for PV curtailment; the reactive-support activation cost is 0.004 per PCC group. Positive PCC active power denotes net import.

DATA SOURCES

[1] xiaoniumang/pandapower, "ieee69bus.txt," 69-bus line and load data. [Online]. Available: https://github.com/xiaoniumang/pandapower/blob/e6e834b2d0312da4994c01d24d5605508bc04fd2/ieee69bus.txt

[2] Open Power System Data, "Household Data," Apr. 15, 2020, DE_KN_residential3. [Online]. Available: https://data.open-power-system-data.org/household_data/2020-04-15/